# Caching and stale information

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

## Before you start

Recall Data retention and deletion. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

Plan time to read, investigate and explain. Keep predictions separate from observations.



## Your mission

> Start with the target: Explain the mechanism and terminology of caching and stale information. Try the recall before opening notes.

### Your learning target

I can explain the **mechanism** and terminology of caching and stale information.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

### Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Data retention and deletion. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Untrusted **input** boundaries. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Approval gates and idempotency. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Explain the **mechanism** and terminology of caching and stale information

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

### Retrieve what you already know

Recall Data retention and deletion. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

### This week’s end goal

Explain caching and stale information using the deciding step in this fictional service.

## Notice the evidence

> Investigate the supplied Caching and stale information case. Separate observations, given facts and assumptions before choosing an explanation.

Caching reuses results but needs freshness and access-aware keys. A cached answer must not cross role or corpus-version boundaries.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

> **List the given facts and one fact that is not supplied.**

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## See how it works

> Follow the mechanism for Caching and stale information. Connect each labelled input, deciding step and result.

**Cache key** includes query, role, corpus version and procedure version. Entry valid only while age<TTL. At exact TTL it expires. Teacher and student results never share a role-agnostic key.

### Caching and stale information — evidence trace

| INPUT AND SCOPE | CHECK OR CALCULATION | CONCLUSION AND LIMIT |
| --- | --- | --- |
| Teacher response is cached by query only and returned to a student. | Cache key includes query, role, corpus version and procedure version. Entry valid only while age<TTL. At exact TTL it expires. Teacher and student results never share a role-agnostic key. | Role omission creates an access leak. Include role and enforce current permission before returning any cached content. |

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

### Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

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STUDENT LESSON · 4 OF 18

## Compare the cases

> Compare the supplied cases for Caching and stale information. Identify the detail that changes the conclusion.

### Compare a familiar case with a changed case

Teacher response is cached by query only and returned to a student. Show the intermediate steps and give the conclusion with its scope.

**Keep the procedure fixed. Explain what changed in the inputs and result.**

### Change one thing in the pictured case

#### Change one thing

Change: The next request still uses approved version 5.

#### Predicted result

**Prediction**: The matching version-5 entry can supply 09:30 under the freshness policy.

#### Why the result follows

Reason: Request and cached source version now agree.

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STUDENT LESSON · 5 OF 18

## Words that unlock the idea

Use the vocabulary of Caching and stale information to explain a concrete decision, rather than only memorising definitions.

| Word | Meaning |
| --- | --- |
| Cache key | Fields identifying a reusable result |
| TTL | The lifetime allowed by the cache contract |
| Invalidation | Marking an entry unusable after a relevant change |

**Use two terms to explain a step in this week's example.**

STUDENT LESSON · 6 OF 18

## Follow a worked example

> Follow the worked trace for Caching and stale information, predict a step, then compare with its explanation.

Teacher response is cached by query only and returned to a student.

Role omission creates an access leak. Include role and enforce current permission before returning any cached content.

### Worked explanation · Caching and stale information

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

### Worked explanation · Caching and stale information

Optional local lab: predict the printed result before running this standalone Python 3 example. Change one relevant boundary **input**, record the outcome and state the contract or statistical assumptions.

The program uses fictional values and the Python standard library; it performs no external operations.

```
def cache_key(query, role, corpus_version, procedure_version):
    return (query, role, corpus_version, procedure_version)

def cache_valid(age, ttl, saved_version, current_version):
    return 0 <= age < ttl and saved_version == current_version

print(cache_valid(10, 10, "v1", "v1"))
```

#### Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

#### A pictorial worked case: Caching and stale information

Cache by source version. A service has an opening-time answer cached from version 4, but version 5 is now approved.

#### The facts we will use

### Supplied facts

Version 4 says 09:00; version 5 says 09:30. **Cache key** contains room and source version.

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# Caching and stale information

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.



## 1. Read authority

The current approved source version is 5.



## 2. Inspect cached key

The 09:00 entry belongs to version 4.



## 3. Fetch current evidence

Retrieve and cache version 5's 09:30 under its own key.

**CHANGE ONE THING**

The next request still uses approved version 5.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

### Read the picture one step at a time

| STEP | WHAT HAPPENS AND WHY |
| --- | --- |
| 1. Read authority | The current approved source version is 5. |
| 2. Inspect cached key | The 09:00 entry belongs to version 4. |
| 3. Fetch current evidence | Retrieve and cache version 5's 09:30 under its own key. |

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows

#### Why the result follows

A cache saves repeated work but can retain obsolete facts. Version-aware keys distinguish answers grounded in different records.

### What this example does not establish

#### Keep the stated limit

Version keys cannot detect silent unversioned edits; the source publication **process** must update versions reliably.

### Keep the stated limit

Read the labelled picture: Identify the version mismatch in the cached 09:00 answer. What should the first version-5 request return, and what permits a later version-5 cache hit?

### Keep the stated limit

Change one thing: The next request still uses approved version 5. Predict the result before checking the explanation. What remains the same?

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STUDENT LESSON · 7 OF 18

## Find and repair a mistake

> Find the unsupported step in the Caching and stale information claim and explain a repair using the supplied evidence.

A cached result retains the limitations of its original **evidence** and can become stale independently of TTL.

### Check the claim against the facts

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

**Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

### Locate the earliest unsupported step

A report omits this qualification: A cached result retains the limitations of its original **evidence** and can become stale independently of TTL.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

**State one repair and the checks you would rerun.**

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STUDENT LESSON · 8 OF 18

## Practise together

> Practise Caching and stale information with support, then explain the deciding step in your own words.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

**Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

**Explain your prediction to a partner before comparing results.**

### Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Caching and stale information”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

| Plan | Do | Check |
| --- | --- | --- |
| What is given? Which rule or procedure applies? | Record each intermediate step. | Does the result follow? What remains unknown? |

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

### Check your reasoning

Use the worked case for Caching and stale information. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

### Explain the picture

> Identify the version mismatch in the cached 09:00 answer. What should the first version-5 request return, and what permits a later version-5 cache hit?

**My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

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STUDENT LESSON · 9 OF 18

## Run the investigation

> Investigate Caching and stale information. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

### Predict, investigate and preserve the evidence

Teacher response is cached by query only and returned to a student. Show the intermediate steps and give the conclusion with its scope.

```
def cache_key(query, role, corpus_version, procedure_version):
    return (query, role, corpus_version, procedure_version)

def cache_valid(age, ttl, saved_version, current_version):
    return 0 <= age < ttl and saved_version == current_version

print(cache_valid(10, 10, "v1", "v1"))
```

| Record | Your evidence |
| --- | --- |
| Given input and fixed rule | |
| Prediction before checking | |
| Actual result, or “not run” | |
| First mismatch and one explanation | |
| Boundary or missing input and expected response | |

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

**Plan → try → check**

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

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STUDENT LESSON · 10 OF 18

## Build your understanding

> Move from recall to tracing, diagnosis and transfer in Caching and stale information. Explain why each step is justified.

**Explain → apply → challenge**

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

A report omits this qualification: A cached result retains the limitations of its original **evidence** and can become stale independently of TTL.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

> **Explain which evidence supports your conclusion and which claim remains unproved.**

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STUDENT LESSON · 11 OF 18

## Connect your project

> Apply this week's idea to Grounded Learning Service. Keep an annotated decision and the evidence supporting it.

Add a labelled caching and stale information **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of caching and stale information.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

| Project evidence | Record |
| --- | --- |
| Component and version | |
| Inputs permitted at prediction time | |
| Trace and comparison with earlier work | |
| Failure, limit and human review | |

### Evidence for your project

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

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STUDENT LESSON · 12 OF 18

## Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

**My limited claim and a useful next check**

STUDENT LESSON · 13 OF 18

# Study a complete case

Revisit the complete Caching and stale information case for supported practice. Repeated examples do not become fresh assessments.

## Complete evidence record

### Caching and stale information — evidence trace

| INPUT AND SCOPE | CHECK OR CALCULATION | CONCLUSION AND LIMIT |
| --- | --- | --- |
| Teacher response is cached by query only and returned to a student. | Cache key includes query, role, corpus version and procedure version. Entry valid only while age<TTL. At exact TTL it expires. Teacher and student results never share a role-agnostic key. | Role omission creates an access leak. Include role and enforce current permission before returning any cached content. |

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

**My complete trace of the worked case**

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STUDENT LESSON · 14 OF 18

## Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

### Independent case: Apply a new case independently

TTL10 minutes; ages9, 10, 11. Corpus version changes at age9. Decide validity.

**My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

### Attempt this without hints

TTL10 minutes; ages9, 10, 11. Corpus version changes at age9. Decide validity.

**My independent procedure, result and limitation**

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

### Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

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STUDENT LESSON · 15 OF 18

## Your lesson route

> Use this map to navigate Caching and stale information, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Cache by source version. A service has an opening-time answer cached from version 4, but version 5 is now approved.

| By the end, I can | How I will show it |
|---|---|
| Explain the mechanism and terminology of caching and stale information | Explain the deciding step, show a trace or test, and state one limit. |
| Apply the stated procedure to a new case with a traceable result | Explain the deciding step, show a trace or test, and state one limit. |
| Identify an assumption or limitation and design a revealing follow-up | Explain the deciding step, show a trace or test, and state one limit. |

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

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STUDENT LESSON · 16 OF 18

## Finish and return

> Explain Caching and stale information without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of caching and stale information

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

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STUDENT LESSON · 17 OF 18

## Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building

> block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Cache by source version. A service has an opening-time answer cached from version 4, but version 5 is now approved.

**Practice 1: explain the deciding evidence**

Which labelled step matches this detail? Retrieve and cache version 5's 09:30 under its own key.

1. Read authority

2. Fetch current **evidence**

3. Inspect cached key

**My selected conclusion, deciding fact and explanation**

**Hint · try first**

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: Fetch current **evidence** Retrieve and cache version 5's 09:30 under its own key. A cache saves repeated work but can retain obsolete facts.

Version-aware keys distinguish answers grounded in different records.

**Practice 2: explain the deciding evidence**

### Changed case · predict first

Changed-case practice: The next request still uses approved version 5. Which conclusion is supported by the supplied facts?

1. The matching version-5 entry can supply 09:30 under the freshness policy.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: The matching version-5 entry can supply 09:30 under the freshness policy. Request and cached source version now agree.

### Repair a missing building block

Review Data retention and deletion (ai-6/20). Goal: Explain the **mechanism** and terminology of data retention and deletion. Short practice: Apply the seven-day boundary to T1 and T2.

How many T2 locations need verification, and may deletion be called complete if its backup remains?. Return to this week and explain what you changed.

Review Untrusted **input** boundaries (ai-6/18). Goal: Explain the **mechanism** and terminology of untrusted **input** boundaries. Short practice: Classify h3, H3, h30 and the trailing-space **input** by the stated shape contract.

Does a valid z0 grant record access, and which separate gate still remains?. Return to this week and explain what you changed.

Review Approval gates and idempotency (ai-6/15). Goal: Explain the **mechanism** and terminology of approval gates and idempotency. Short practice: Trace the two deliveries of approved K9.

How many reservations should occur, and why can a new K10 for desk 5 not borrow K9's approval?. Return to this week and explain what you changed.

### Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

### Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

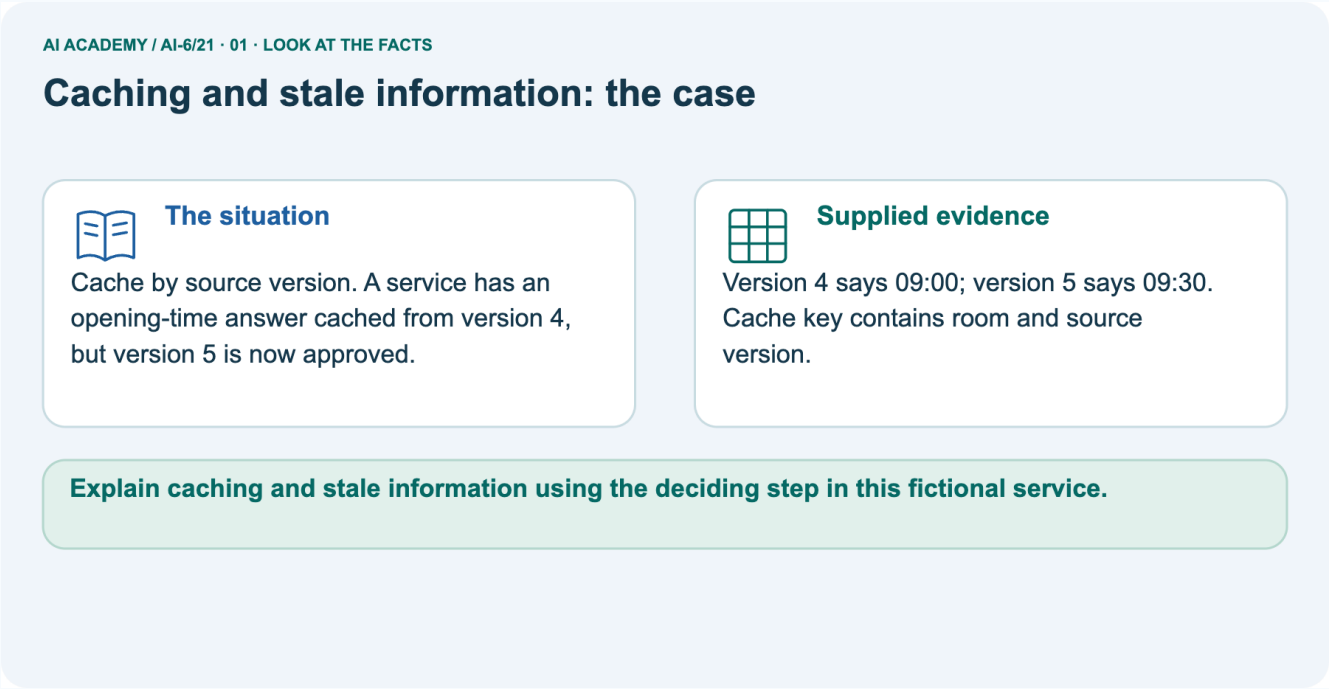
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STUDENT LESSON · 18 OF 18

# Visual concept journey

> Explain each Caching and stale information graphic using its labels, deciding evidence and relevant limitation.

### 01. Caching and stale information: the case

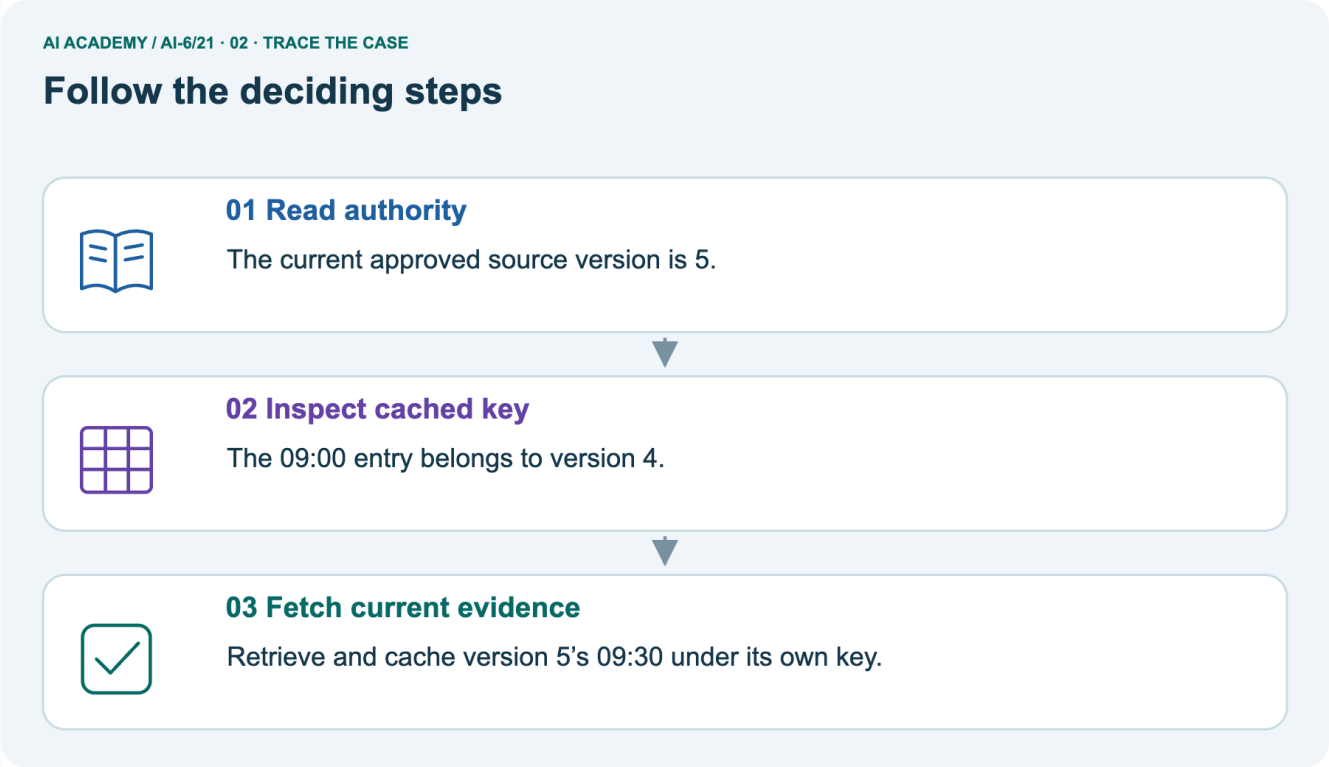



Follow the labelled picture with the matching explanation. Open the vector diagram

Cache by source version. A service has an opening-time answer cached from version 4, but version 5 is now approved. Supplied facts: Version 4 says 09:00; version 5 says 09:30. **Cache key** contains room and source version.

> Point to the given fact that matters most. Explain why it matters.

### 02. Follow the deciding steps

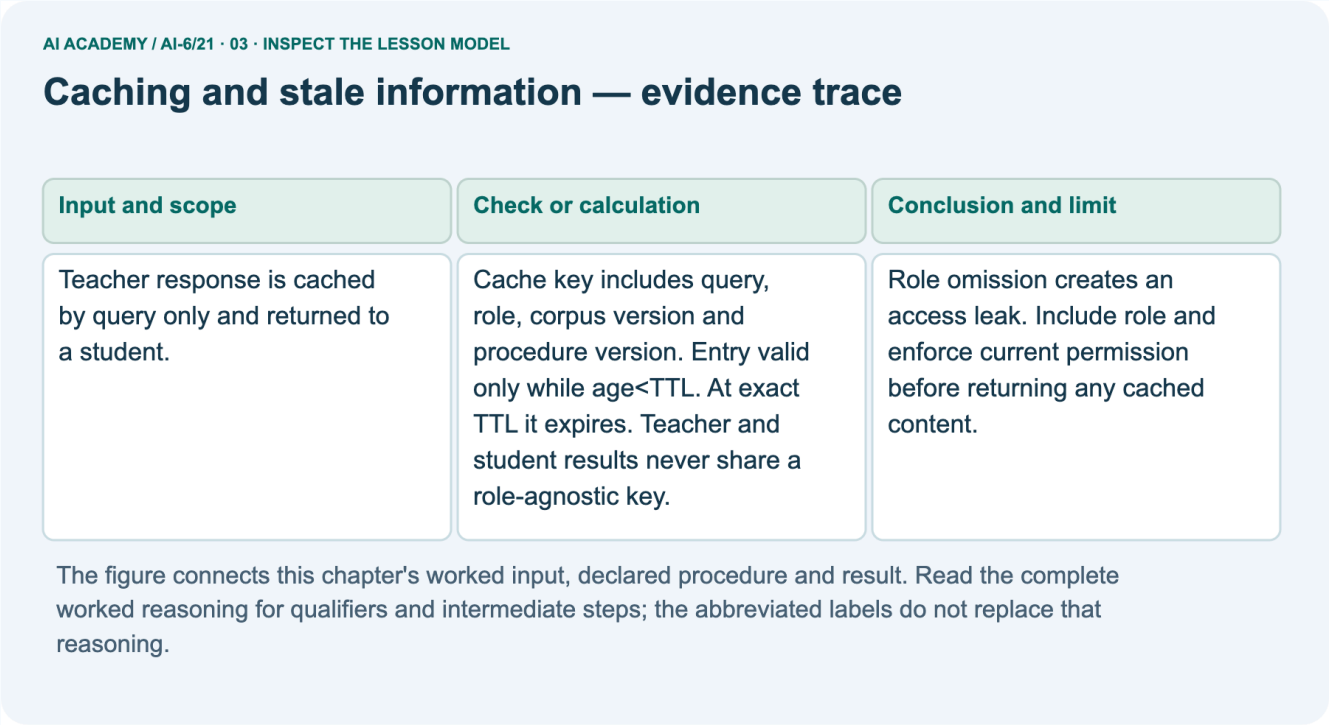



Follow the labelled picture with the matching explanation. Open the vector diagram

Read authority: The current approved source version is 5. → Inspect cached key: The 09:00 entry belongs to version 4. → Fetch current **evidence**: Retrieve and cache version 5's 09:30 under its own key.

> Point to each numbered stage and explain which recorded **input** it uses.

### 03. Caching and stale information — evidence trace



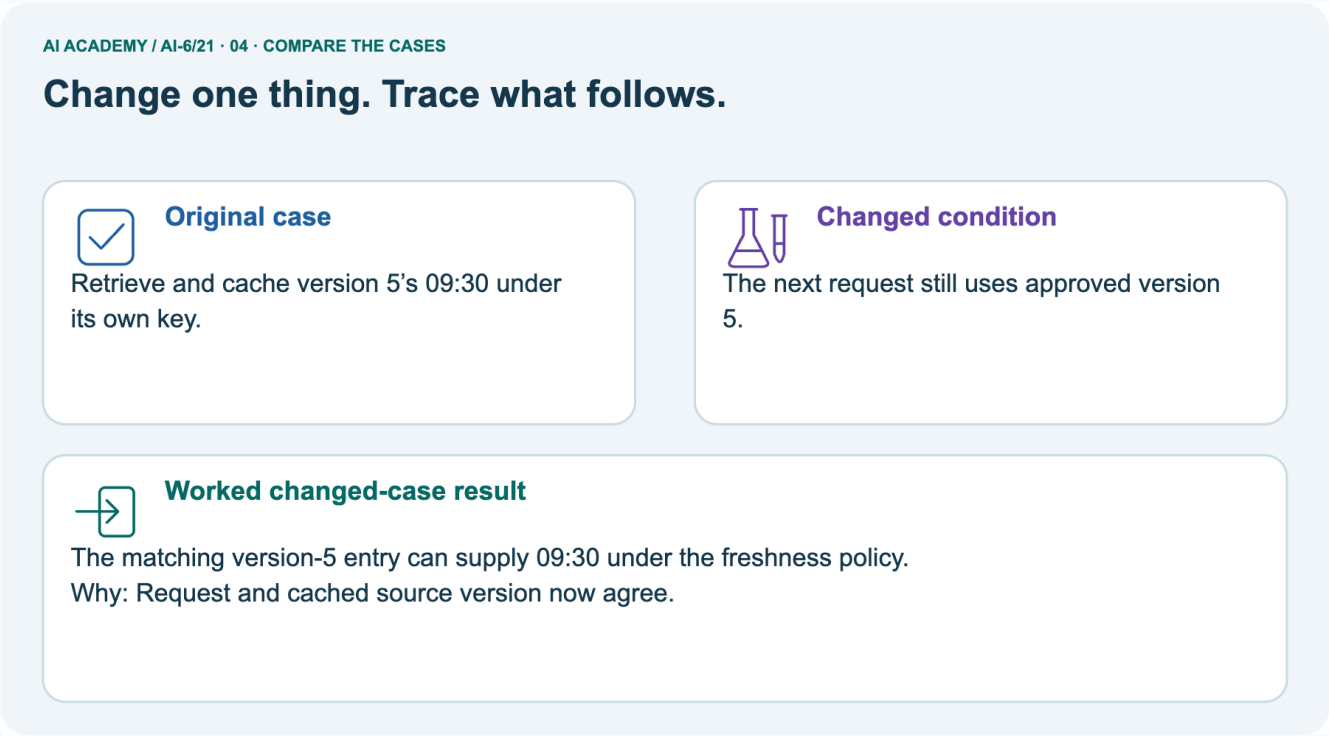

Follow the labelled picture with the matching explanation. Open the vector diagram

Caching and stale information — **evidence** trace. Teacher response is cached by query only and returned to a student. | **Cache key** includes query, role, corpus version and procedure version. Entry valid only while age<TTL.

At exact TTL it expires. Teacher and student results never share a role-agnostic key. | Role omission creates an access leak. Include role and enforce current permission before returning any cached content.

> Read the labels and values. Explain the relationship the model shows.

## 04. Change one thing. Trace what follows.



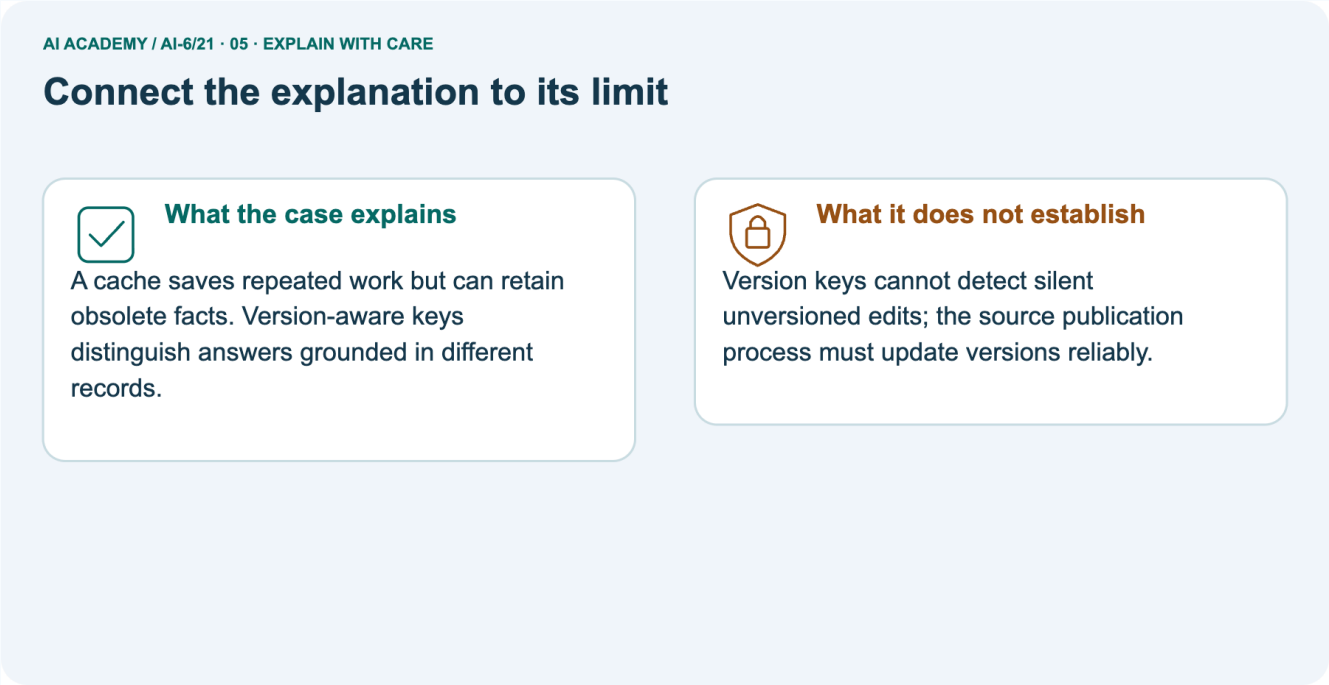

Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Retrieve and cache version 5's 09:30 under its own key. Changed condition: The next request still uses approved version 5.

Expected result: The matching version-5 entry can supply 09:30 under the freshness policy. Reason: Request and cached source version now agree.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

## 05. Connect the explanation to its limit



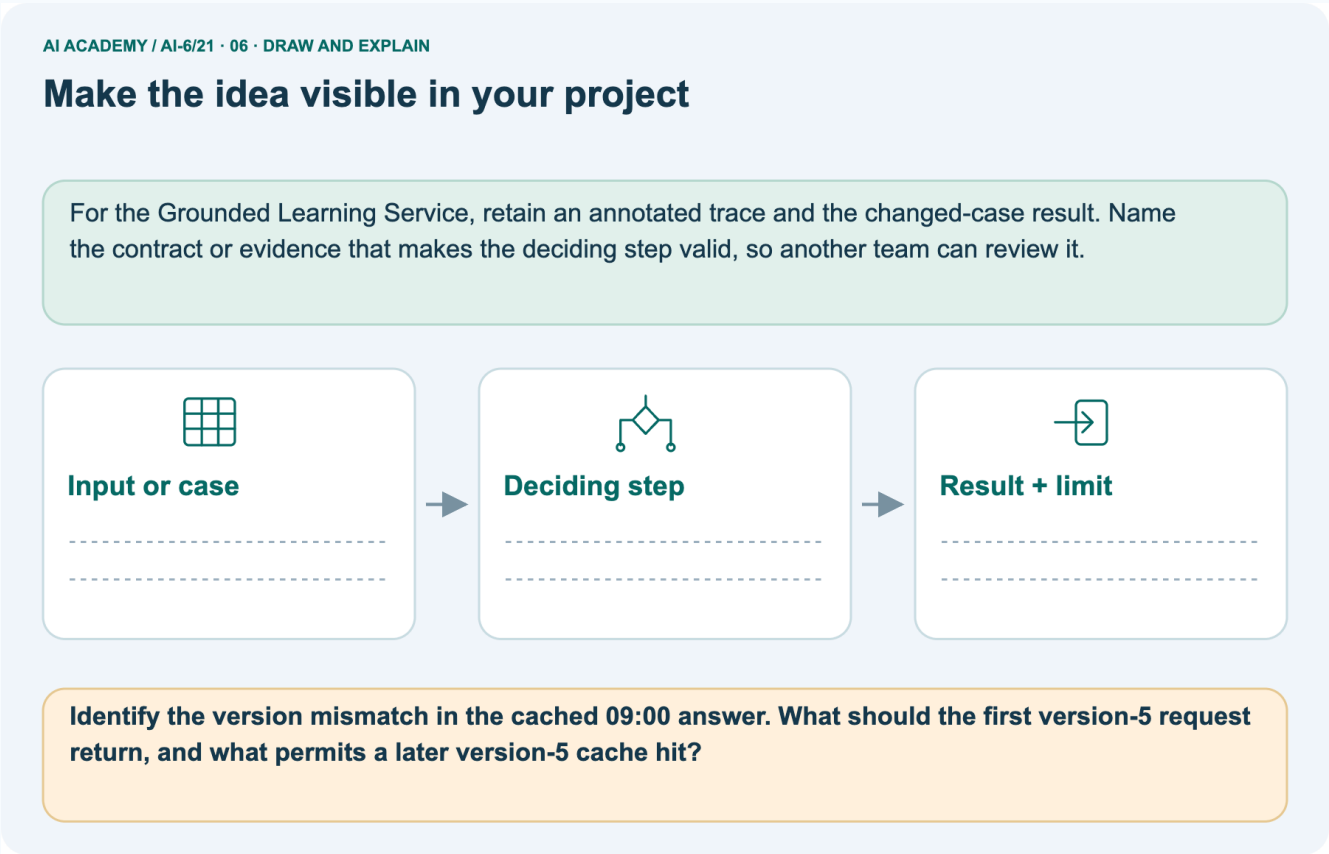

Follow the labelled picture with the matching explanation. Open the vector diagram

A cache saves repeated work but can retain obsolete facts. Version-aware keys distinguish answers grounded in different records.

Limit: Version keys cannot detect silent unversioned edits; the source publication **process** must update versions reliably.

> Explain the case, then point to a limit on the claim.

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Identify the version mismatch in the cached 09:00 answer. What should the first version-5 request return, and what permits a later version-5 cache hit?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.